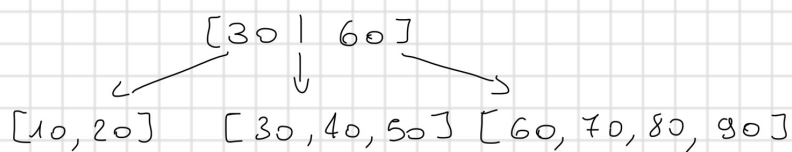


Fen out = 3

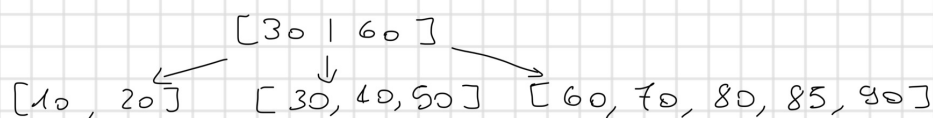
Fen out = 3

- Nodi intermedi: $3 \leq \# \text{puntatori} \leq 5$

↓
(max 4 chiori)



1) Inseriamo la chiave 85 nell'albero di partenza.

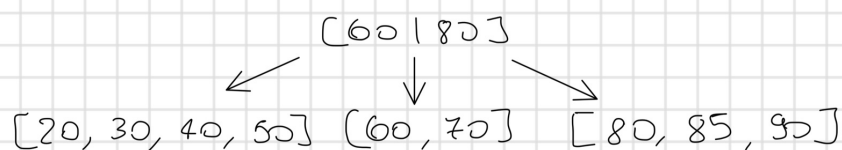


$[30 | 60 | 80]$
 $\swarrow \quad \searrow \quad \searrow \quad \searrow$
 $[10, 20] \quad [30, 40, 50] \quad [60, 70] \quad [80, 85, 90]$

2) Cancella il 10

L'albero non rispetta più i vincoli:

Applichiamo il MERCÉ

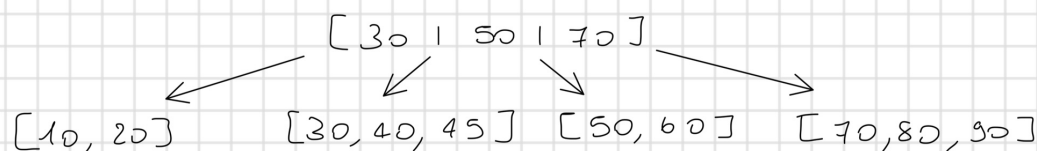


Es 2

$$f_{\text{out}} = 4$$

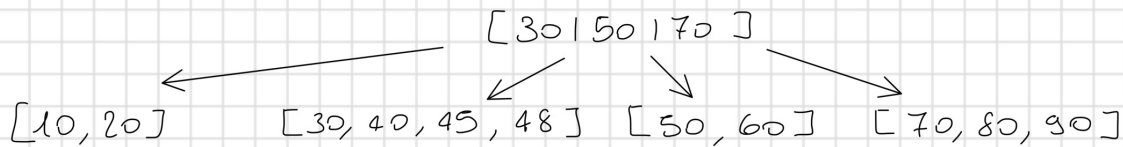
- Nodi intermedi = $2 \leq \# \text{punteggi} \leq 4$

↓
(max: 3)

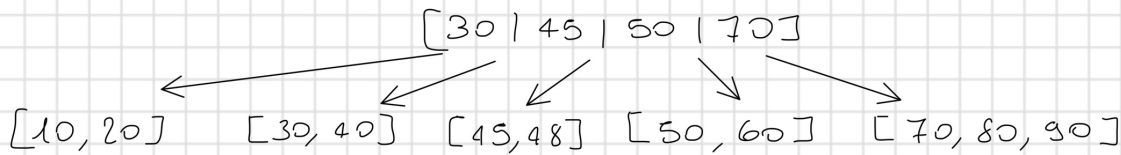


Rispetta i vincoli

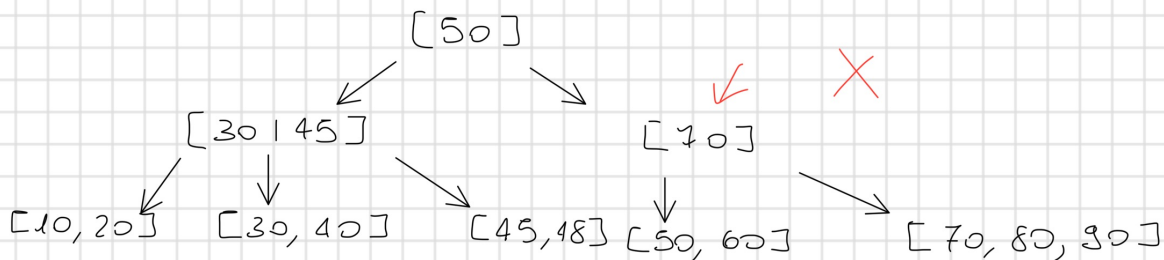
1) Inserimento 48



Non rispetta i vincoli, applichiamo uno SPLIT

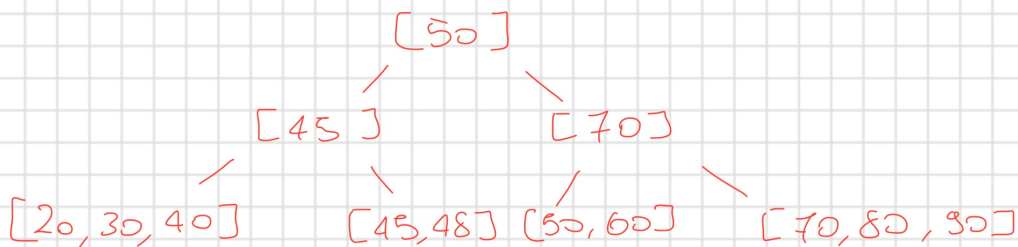
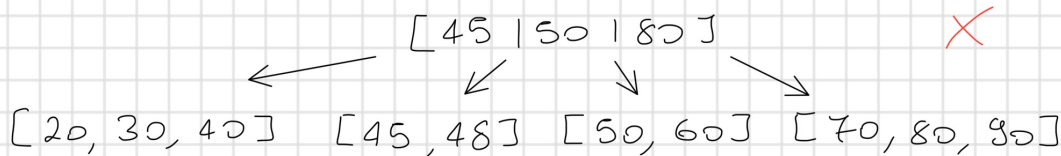


Non rispetta i vincoli, SPLIT del nodo intermedio



Cancella 10:

Non rispetta i vincoli, facciamo MERGE



Es 3

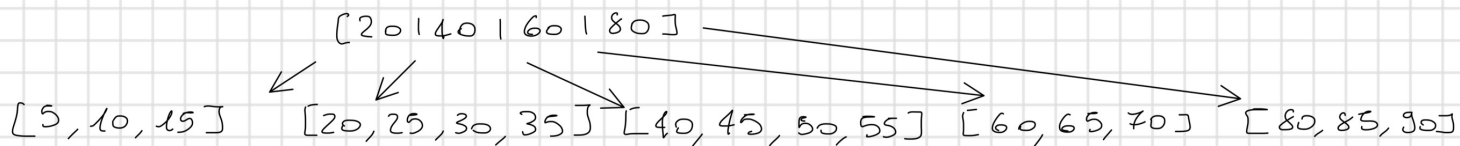
Fan out: 5

Foglie: $2 \leq \# \text{ chiavi} \leq 4$

Intermedio: $3 \leq \# \text{ puntatori} \leq 5$

(min: 2)

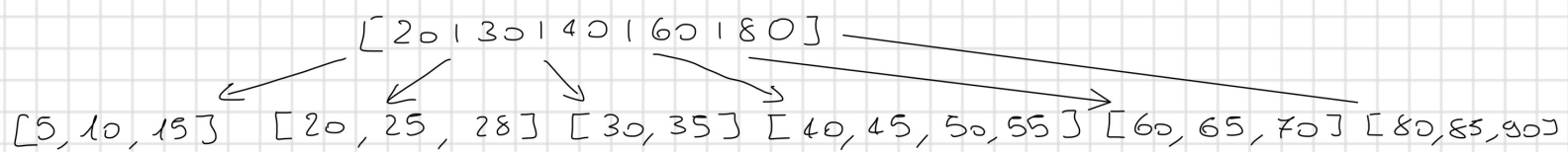
(min: 4)



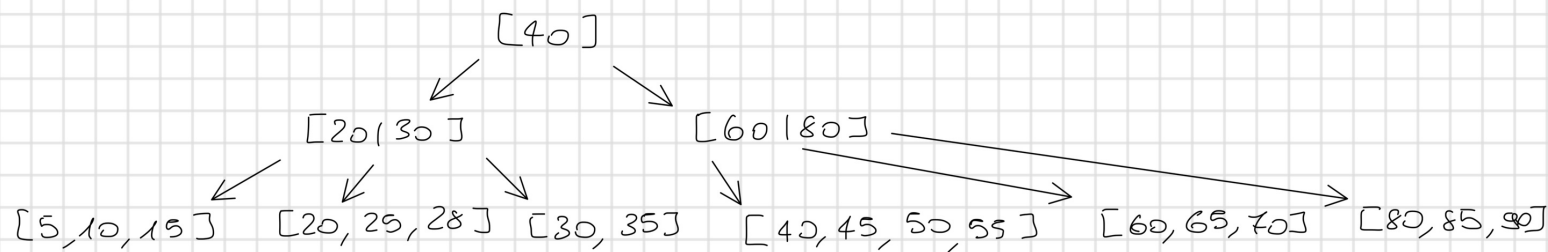
Rispetta i vincoli

1) Aggiungi 28:

Non rispetta i vincoli, dobbiamo fare SPLIT



Non rispetta i vincoli, faccio SPLIT

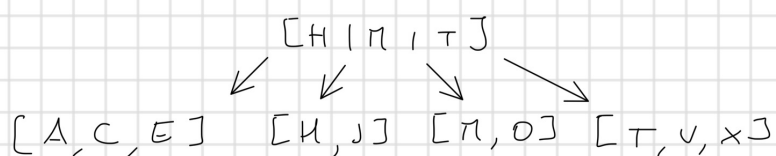


Es 4

Fan out: 4

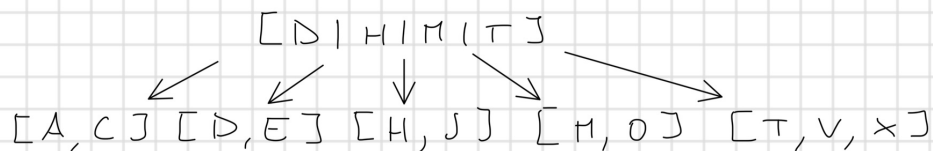
Foglie: $2 \leq \# \text{ chiavi} \leq 3$

Intermedi: $2 \leq \# \text{ puntatori} \leq 4$
 ↓ ↓
 (min: 1) (max: 3)

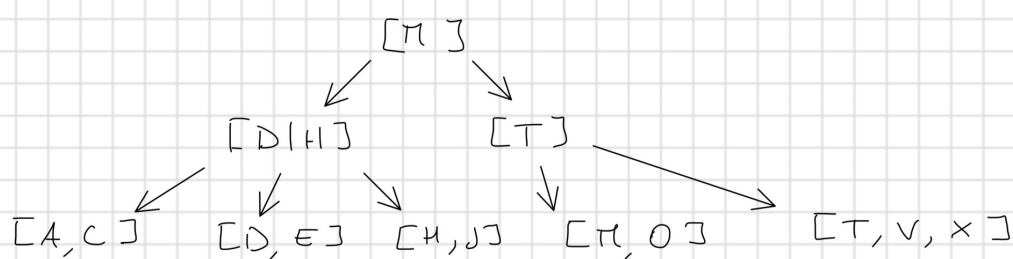


Inserisci D

Non rispetta i vincoli, SPLIT

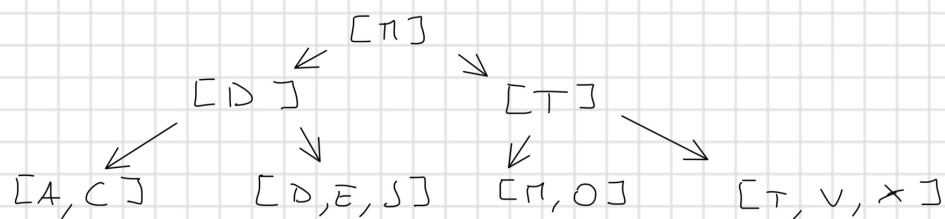


Non rispetta i vincoli, SPLIT



2) Cancella H

Va in underflow, MERGE



Es 5

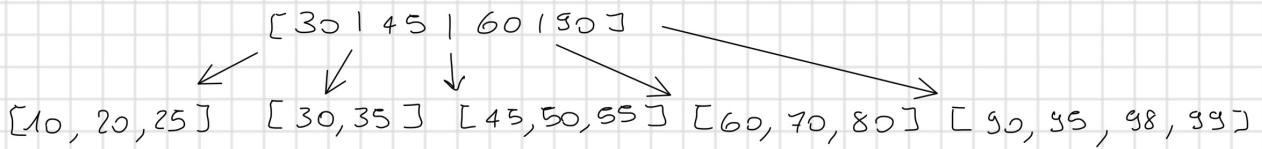
fan out: 5

foglie: $2 \leq \# \text{chiavi} \leq 4$

intermedi: $3 \leq \# \text{punteggi} \leq 5$

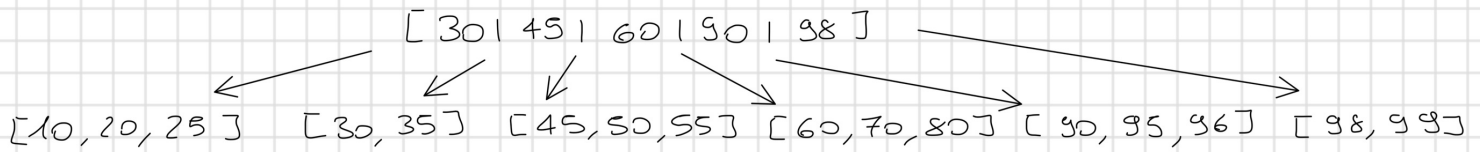
(min: 2)

(max: 4)

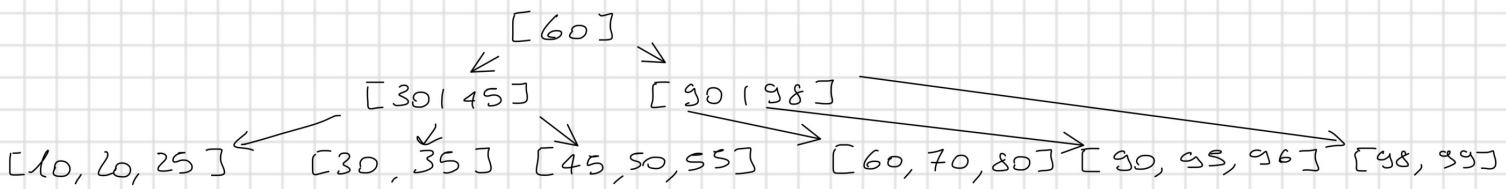


Inserisci 96

Non rispetta i vincoli, SPLIT

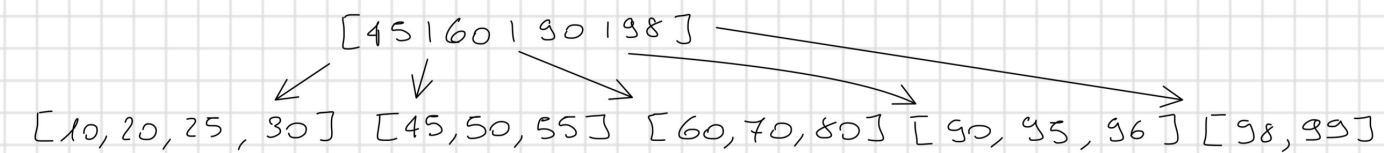


Non rispetta i vincoli, SPLIT



Rimuovi 35

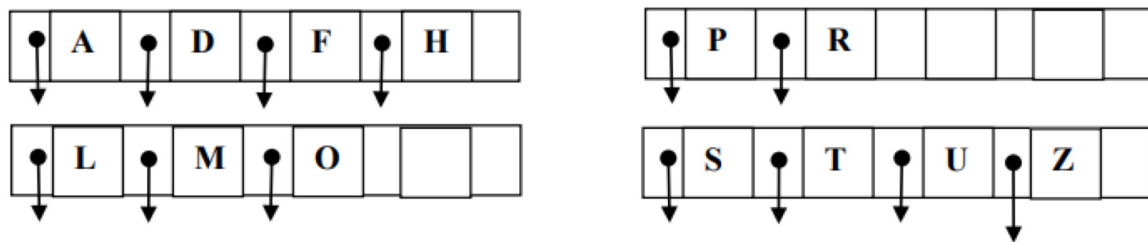
Non rispetta i vincoli, MERGE



B⁺-tree

Data la seguente lista di possibili valori chiave $L = (A, B, C, D, E, F, G, H, I, L, M, N, O, P, Q, R, S, T, U, W, Z)$:

a) (2) costruire un B⁺-tree (fan-out=5) che contenga i seguenti nodi foglia:

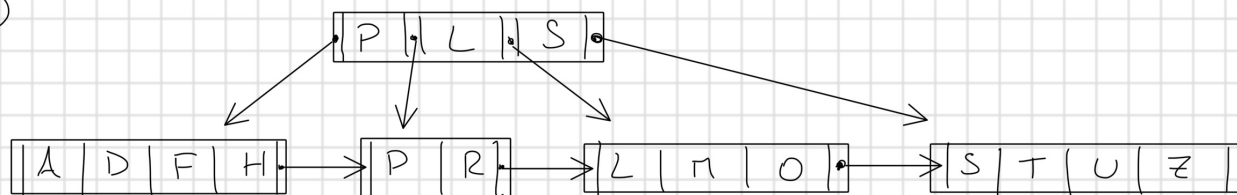


b) (4) mostrare l'albero dopo l'inserimento del valore chiave B, poi del valore C, poi di E, poi di Q.

foglie: $2 \leq \# \text{chiavi} \leq 4$

intermedi: $3 \leq \# \text{puntatori} \leq 5$
 $\downarrow$ (min: 2) $\downarrow$ (max: 4)

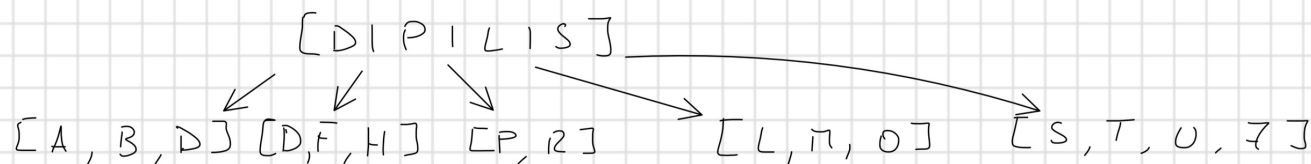
a)



Rispetta i vincoli

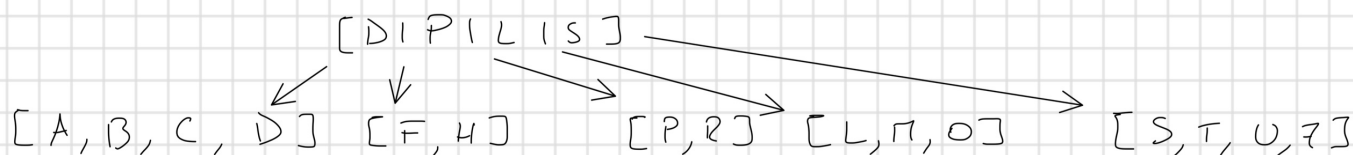
b) Inserisci B

Non rispetta i vincoli, MERGE



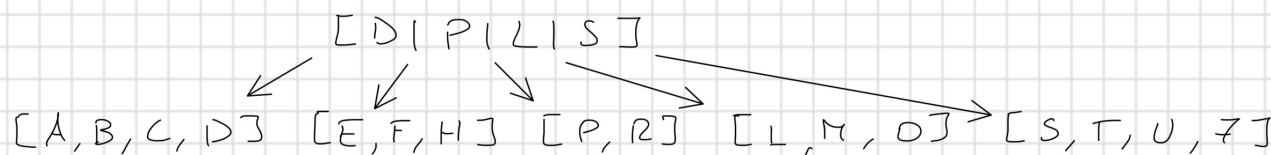
Rispetta i vincoli

Inserisci C

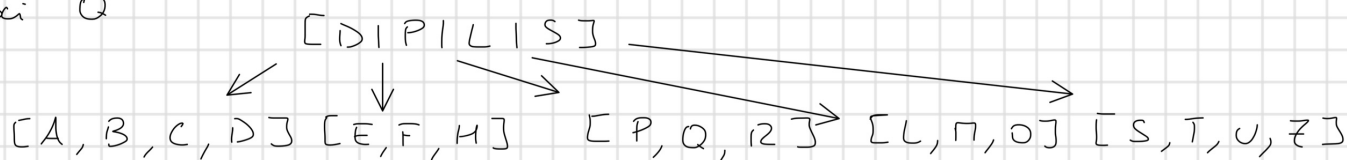


Rispetta i vincoli

Inserisci E



Inserisci Q



Es 7

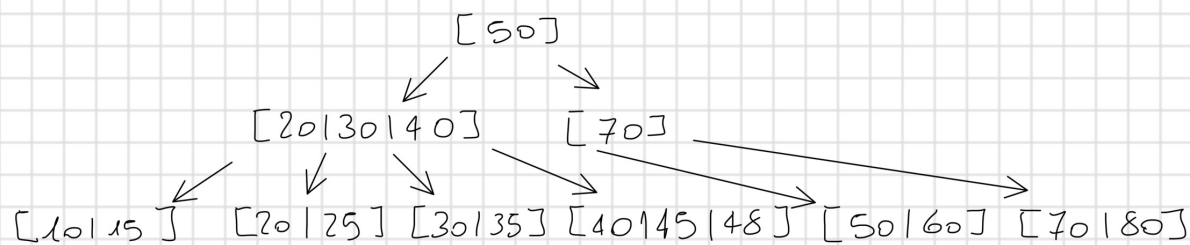
fan out: 4

foglie: $2 \leq \# \text{chiavi} \leq 3$

intermedi: $2 \leq \# \text{punteggi} \leq 4$

↓
(min: 1)

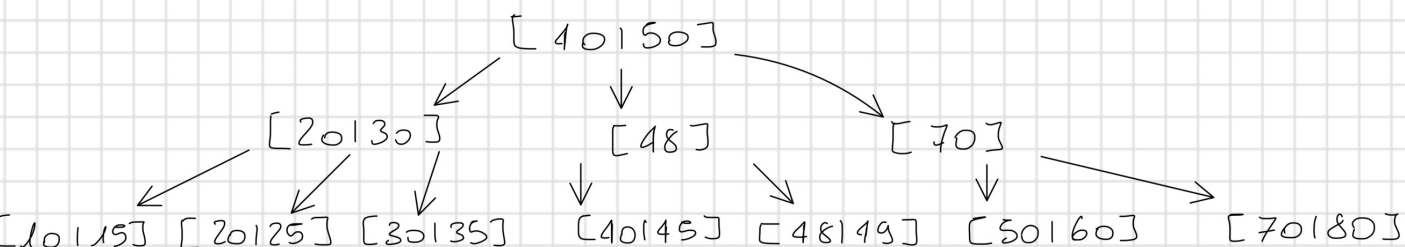
↓
(max: 3)



Rispetta i vincoli

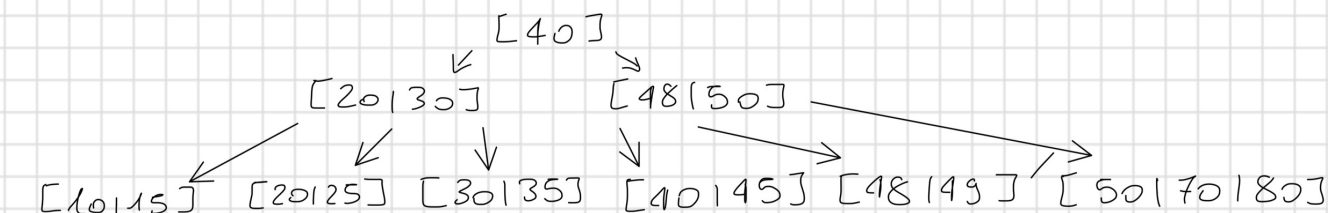
Inserisci 49

Non rispetta i vincoli, SPLIT



Cancella 60

Non rispetta i vincoli, MERGE



Es 8

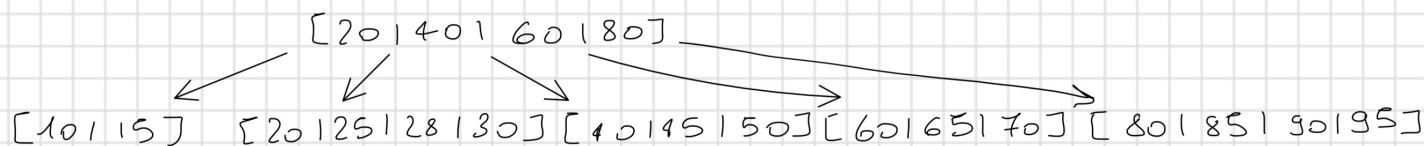
fan out: 5

foglie: $2 \leq \text{chiavi} \leq 4$

intermedi: $3 \leq \text{punteggi} \leq 5$

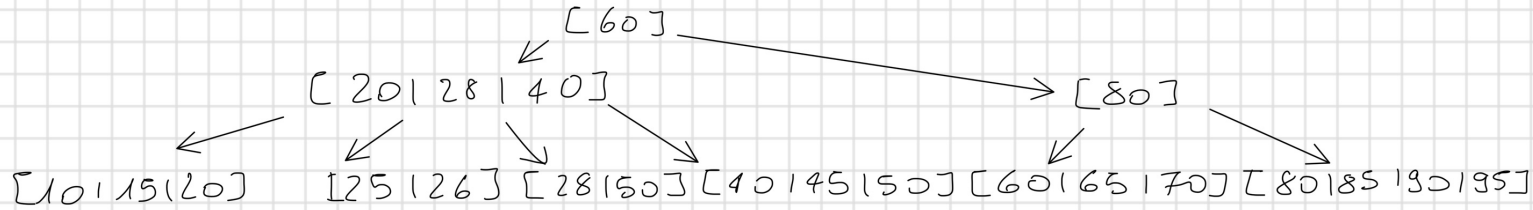
↓
(min: 2)

↓
(max: 4)



Inserisci 26

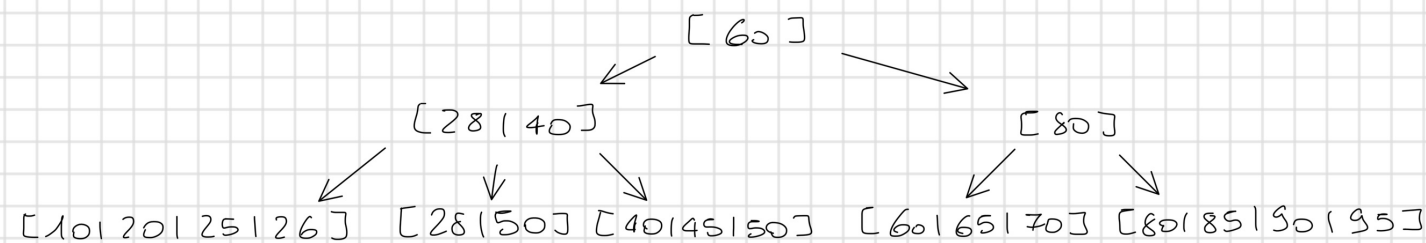
Non rispetta i vincoli, SPLIT



Cancella 15

Non rispetta i vincoli, MERGE

~~X~~ sbagliato, 2 fogli



Es 9 ~~X~~ Ricardoli (l'ordine

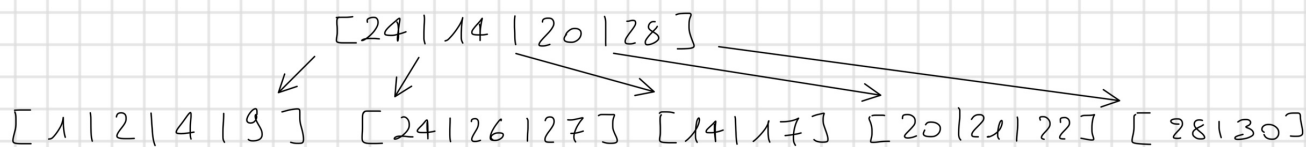
fan out = 5

foglie = $2 \leq \# \text{ chiavi} \leq 4$

intermedi = $3 \leq \# \text{ puntatori} \leq 5$

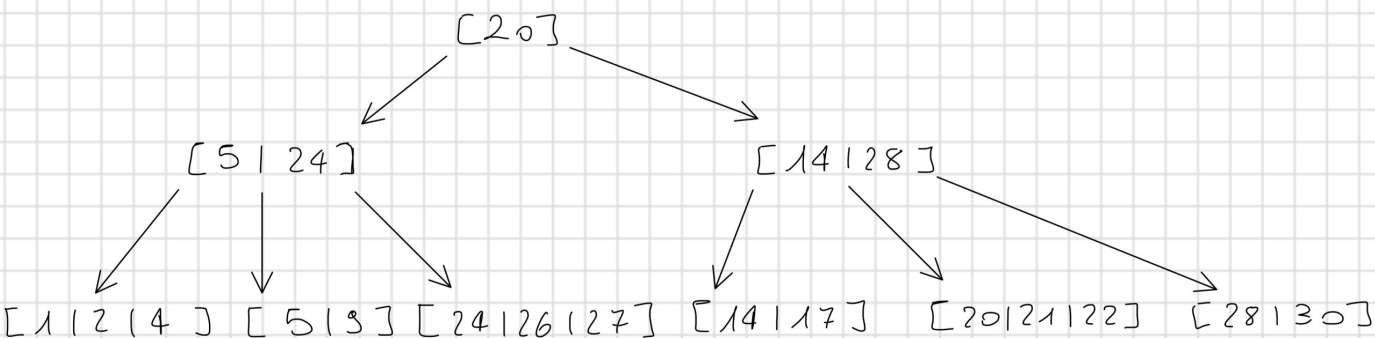
(min: 2)

(max: 4)



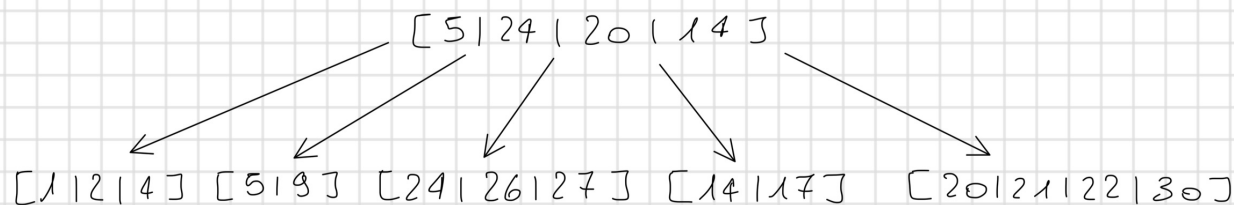
Inserisci 5

Non rispetta i vincoli, SPLIT



Topli 28

Non rispetta i vincoli, MERGE



ES 10

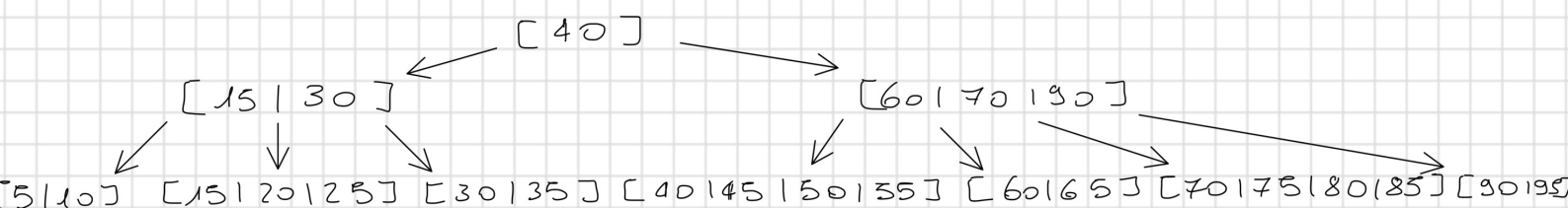
fan out = 5

foglie = $2 \leq \# \text{chiodi} \leq 4$

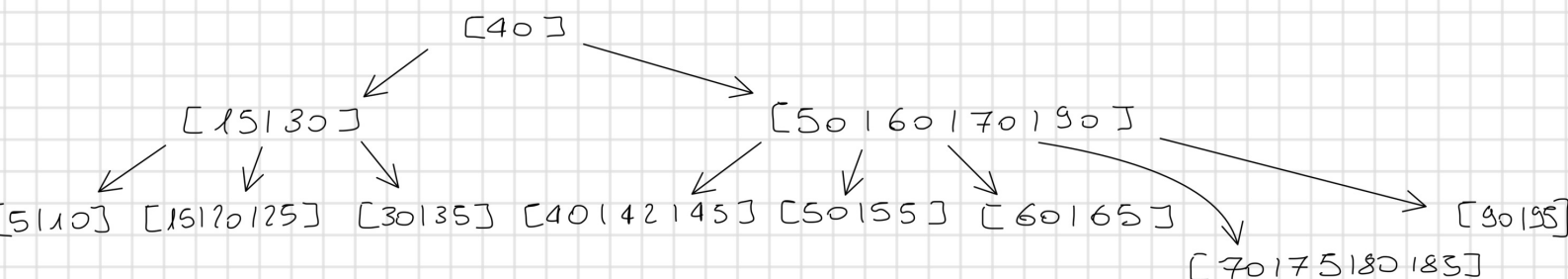
intermedi = $3 \leq \# \text{puntatori} \leq 5$

↓
(min: 2)

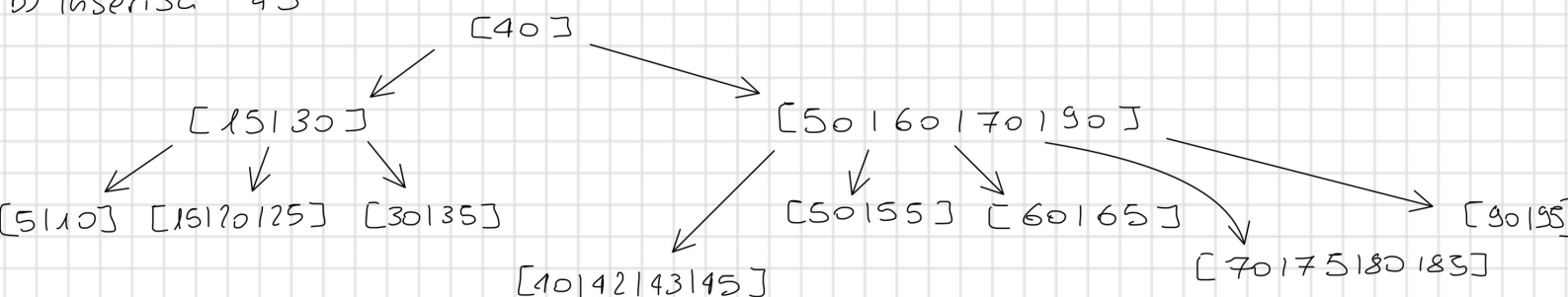
↓
(max: 4)



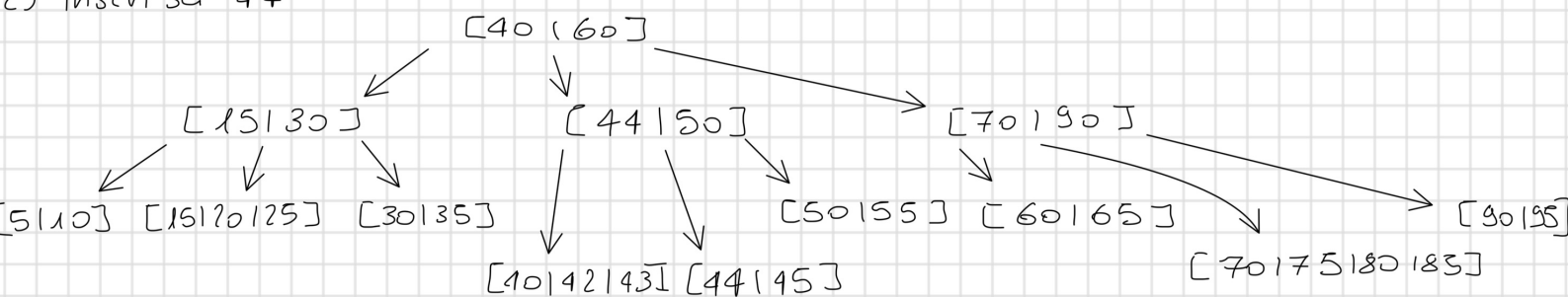
a) Inserisci 42



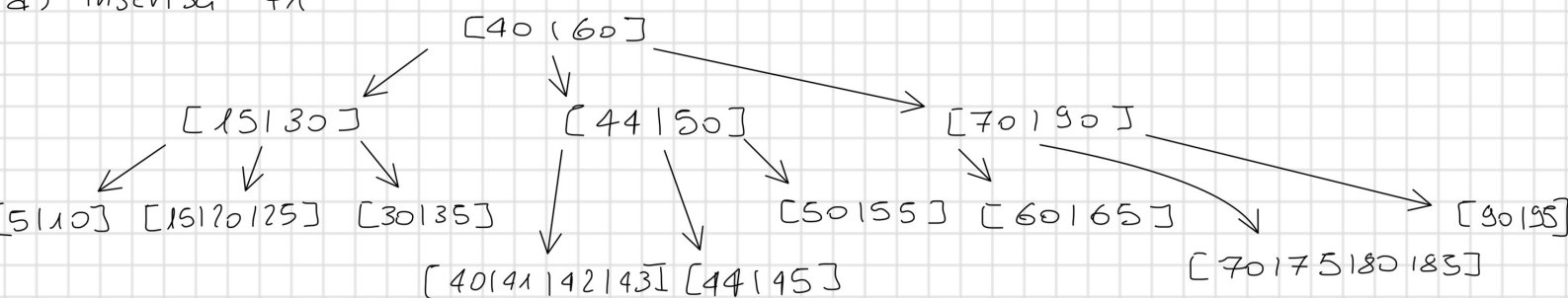
b) Inserisci 43



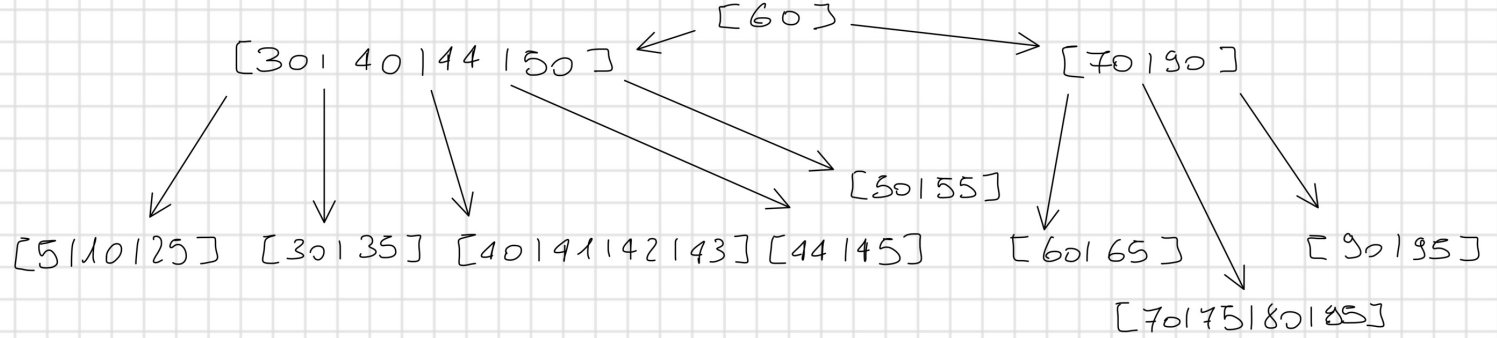
c) Inserisci 44



d) Inserisci 41



e) Rimuovi 15 e 20



Es 11

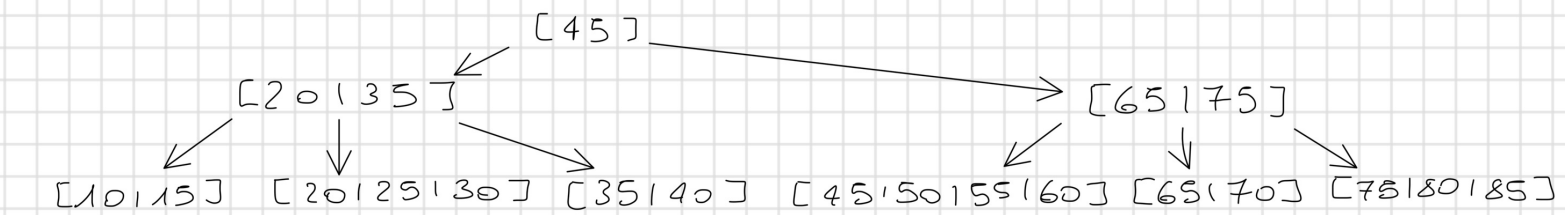
fan out: 5

foglie = 2 ≤ # chiavi ≤ 4

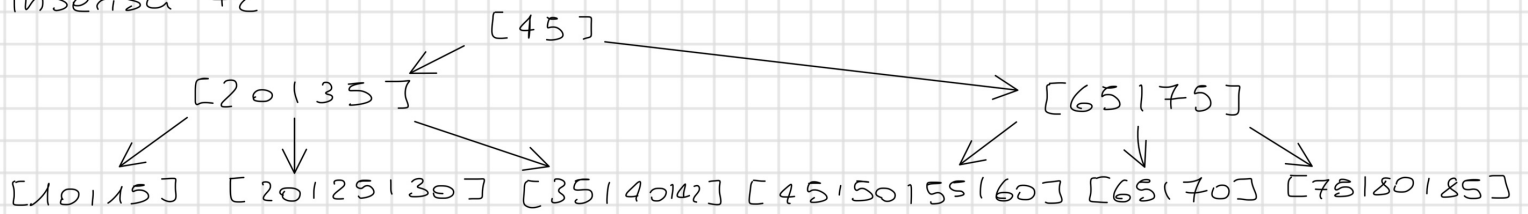
intermedi = 3 ≤ # puntatori ≤ 5

↓
(min: 2)

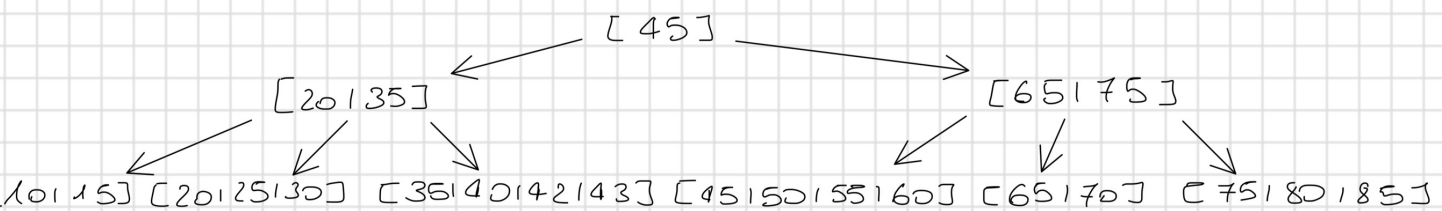
↓
(max: 4)



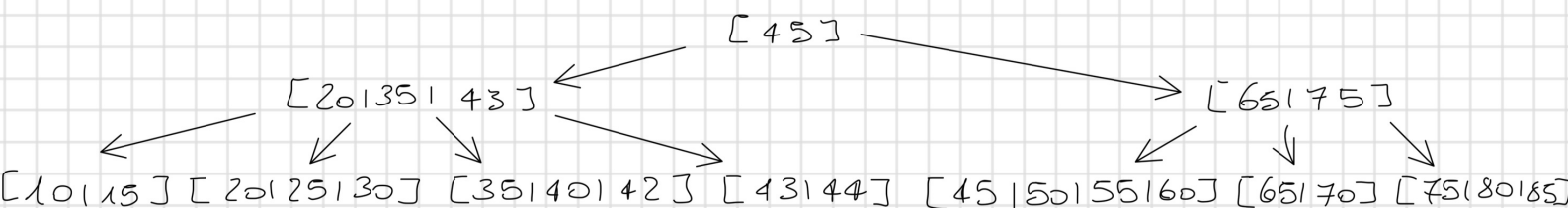
Inserisci 42



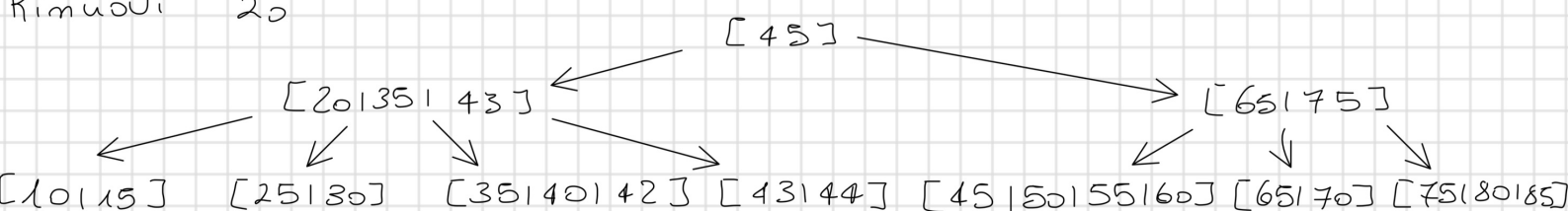
Inserisci 43



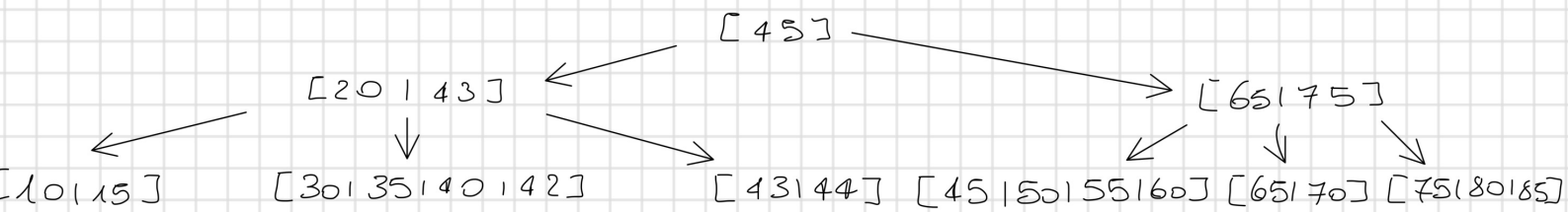
Inserisci 44



Rimuovi 20



Rimuovi 25



Es 1

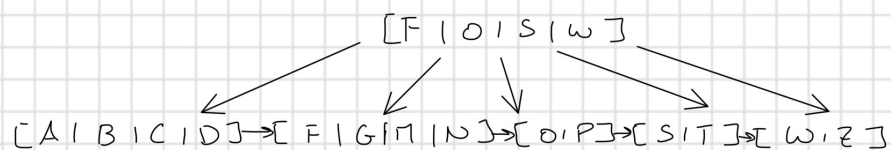
fan out = 5

foglie = $2 \leq \# \text{chiavi} \leq 4$

intermedi = $3 \leq \# \text{puntatori} \leq 5$

↓
(min: 2)

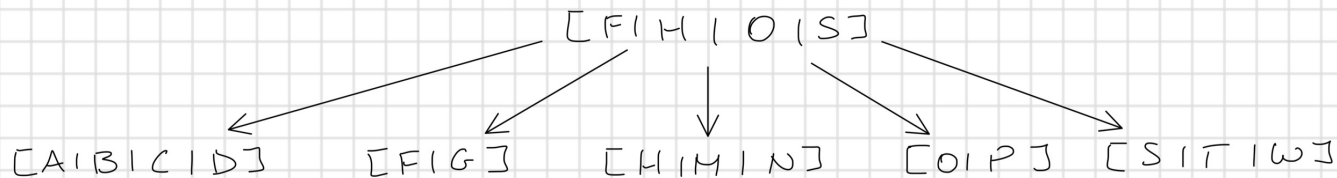
↓
(max: 4)



2) Aggiungiamo H



3) Cancella Z



3)

fan out = 5

foglie = $2 \leq \# \text{chiavi} \leq 4$

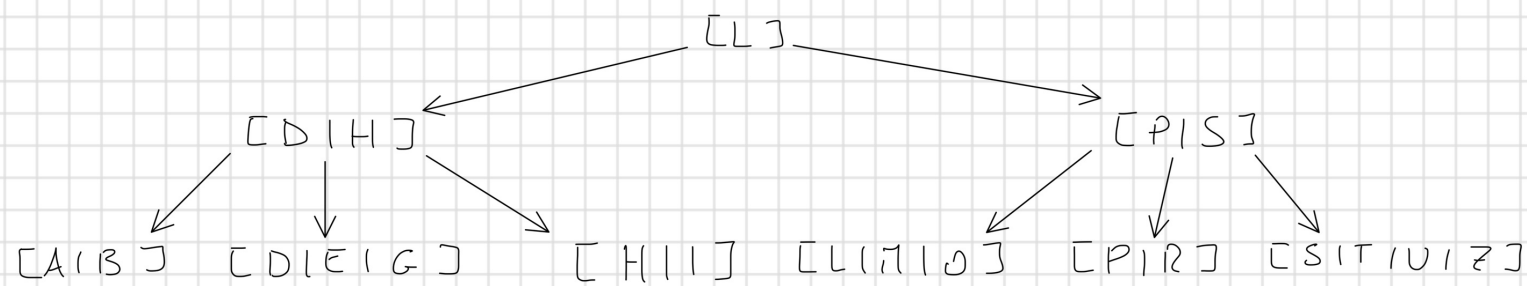
intermedi = $3 \leq \# \text{puntatori} \leq 5$

↓
(min: 2)

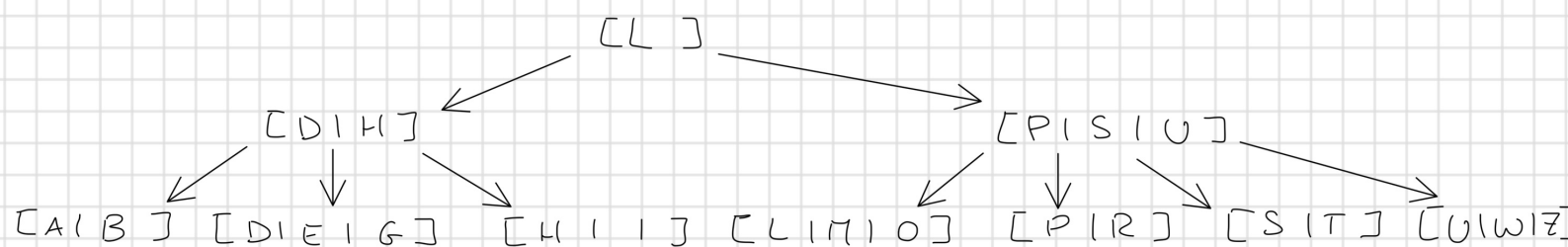
↓
(max: 4)



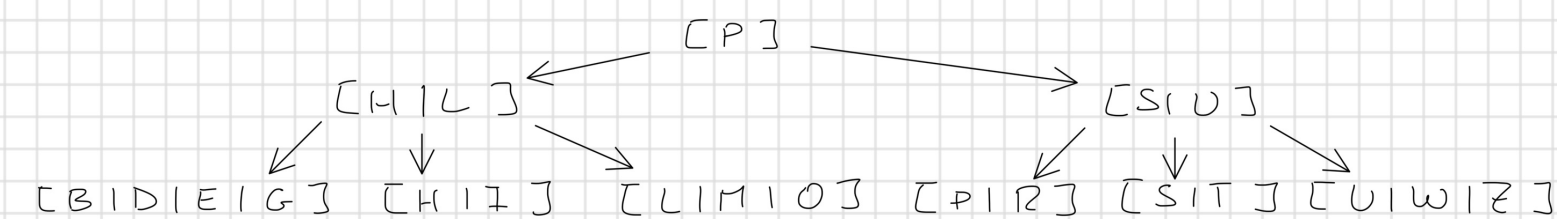
2) Inserisci B



3) Insenisci w



4) CANCELLA A



5) Cancella B

